

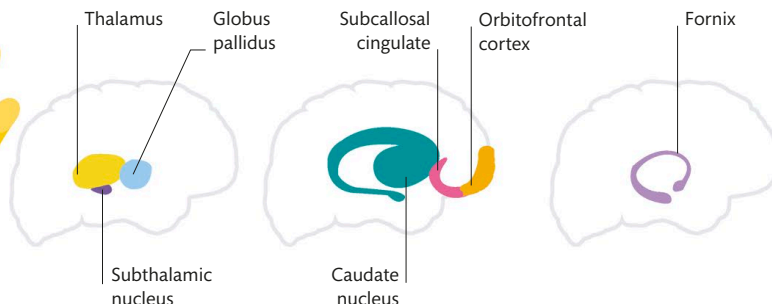


Deep brain stimulation (DBS)

In DBS, electrode wires are implanted in various parts of the brain (see below) to treat a range of disorders. These send pulses of electricity from a generator and battery in the chest, connected to the electrodes. A remote controller adjusts the pulses. In adaptive DBS, the electrodes have sensors and the generator automatically responds to the brain's electrical activity.



THE BATTERIES USED IN PULSE GENERATORS FOR DEEP BRAIN STIMULATION LAST UP TO ABOUT NINE YEARS



Movement disorders

DBS is well established to treat movement problems, such as the tremors and "freezing" of Parkinson's disease and the spasms and contractions of dystonia.

Psychiatric disorders

DBS may be used in severe anxiety, depression, and obsessive-compulsive disorder, where other treatments such as drug medication have not proved effective.

Cognitive disorders

Research explores DBS for problems such as Alzheimer's disease, targeting specific structures involved in memory and cognitive neural networks.

WHEN WAS THE FIRST BIONIC LIMB CREATED?

In 1993, a team of bioengineers at the Margaret Rose Hospital in Edinburgh created the first bionic arm for amputee Robert Campbell Aird.

Vagus nerve stimulation

The vagus nerve, one of the cranial nerves (see p.12) connects the brain with organs in the chest and abdomen. In vagus nerve stimulation (VNS), a small signal generator in the chest, similar to a heart pacemaker, is connected by wires to electrodes around the left vagus nerve in the neck. The nerve's sensory fibers are stimulated to send impulses into the brain, where they are distributed along various neural pathways. VNS is mainly used to treat forms of epilepsy and depression.

